

Unit 8, Lesson 3: Solving Equations with Variables on Both Sides

Benchmark

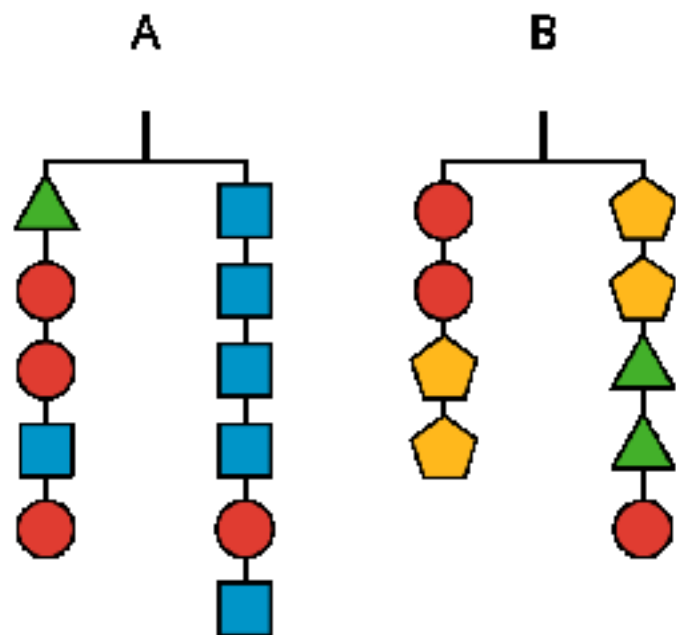
MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

Learning Target

☐ I can solve equations with variables on both sides.

8.3.1 Warm-Up: Balancing Hangers

The picture represents two balanced hangers. The hangers are balanced because the weight on each side is the same.



1. In both hanger diagrams, a triangle weighs 3 grams and a circle weighs 6 grams.

a. Find the weight of a square in Hanger A.

 = _____ grams

b. Write an equation to represent each hanger, where s represents the weight of each square and p represents the weight of each pentagon, in grams.

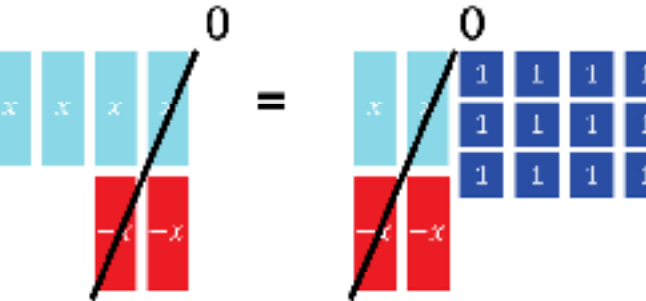
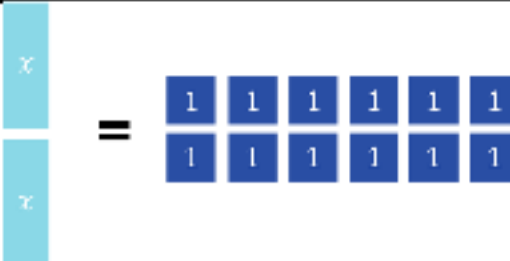
Hanger	Equation
A	
B	

8.3.2 Exploration: Equations with Variables on Both Sides

Similar to hanger diagrams, algebra tiles can be used to model equations. As seen in the Warm-Up, it is possible to have linear equations with variables on both sides of the equal sign.

Work with your partner to use algebra tiles to explore how to solve such equations.

1. Use algebra tiles to solve the equation $4x = 2x + 12$.
- a. Complete the table to describe the process for solving each equation.

Algebra Tile Model	Description	Algebraic Equation
	Given equation	$4x = 2x + 12$
	Add _____ to both sides of the equation to create zero pairs	$4x = 2x + 12$ $-2x \quad -2x$ $2x = 12$
	Divide both sides of the equation into _____ groups	$\frac{2x}{2} = \frac{12}{2}$

b. What is the solution to $4x = 2x + 12$ based on the work done in the table in part a?

2. Use algebra tiles to solve the equation $8x + 15 = 4x - 1$.

a. Complete the table to describe the process for solving each equation.

Algebra Tile Model	Description	Algebraic Equation
	Given equation	$8x + 15 = 4x - 1$
	Add _____ to both sides of the equation to create zero pairs	$8x + 15 = 4x - 1$ $4x + 15 = -1$
	Add _____ to both sides of the equation to create zero pairs	$4x + 15 = -1$ $-15 \quad -15$ $4x = \square$
	Divide both sides of the equation into _____ groups	$\frac{4x}{4} = \frac{\square}{4}$

b. What is the solution to $8x + 15 = 4x - 1$, based on the work done in the table in part a?

3. Use algebra tiles to solve the equation $6 - 3x = 8 + (-4x) + 2$.

a. Use the process of solving the equation with algebra tiles to complete the algebraic representation of the process.

Algebra Tile Model	Algebraic Representation
	$6 - 3x = 8 + (-4x) + 2$
	$6 - 3x = 8 + (-4x) + 2$ $\square \quad \square$ $6 = 8 - x + 2$
	$6 - 3x = 8 + (-4x) + 2$ $\square \quad \square$ $6 = 8 - x + 2$
	$6 = 8 - x + 2$ $6 = \square - x$

1

1

1

1

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-1

-1

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-1

-1

-1

=

-1

-1

-1

-1

-1

-x

6

=

-x

-4

=

-x

- b. Marcus said both sides of the last equation in the table should be either multiplied or divided by -1 . Explain whether you agree with Marcus' recommendation.
- c. Based on your work in Parts A and B, what is the solution to $6 - 3x = 8 + (-4x) + 2$?

4. Discuss with your partner the similarities and differences between solving equations with variables on only one side and solving equations with variables on both sides. Record similarities and differences discussed.

Similarities	Differences

8.3.3 Guided Instruction: Solving Equations with Variables on Both Sides

1. Solve the equation $3y - 4 = 6 - 2y$ by following the steps described in the table.

Description	Algebraic Representation
Begin with the given equation.	$3y - 4 = 6 - 2y$
Eliminate the constant term from one side of the equation using zero pairs.	
Eliminate the variable term from one side of the equation using zero pairs.	
Isolate the variable.	

For some multi-step linear equations, it may be helpful to apply the distributive property or combine like terms on one side of the equation first.

2. Solve the equation $4(m + 7.2) = -2m - 8.1$ by following the steps described in the table.

Description	Algebraic Representation
Begin with the given equation.	$4(m + 7.2) = -2m - 8.1$
Apply the distributive property.	
Eliminate the variable term from one side of the equation.	
Eliminate the constant term from one side of the equation.	
Isolate the variable.	

3. Solve the equation $-12p + 3p - 11 = 14 - 3p - 7$.

8.3.4 Your Turn!

Solve each equation. Verify your solution using substitution.

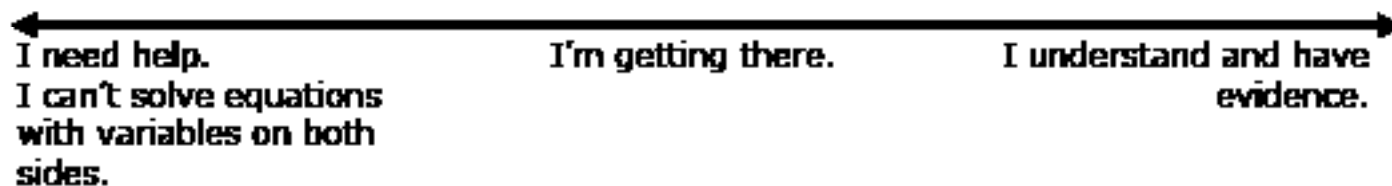
1. $6 - x = 5x + 30$

2. $9(p - 4) = 3(2p + 6)$

8.3.5 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate how you feel about today's learning target.

I can solve equations with variables on both sides.



2. Explain your placement of "X" on the scale.